

DEREK WANG

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Education

University of Waterloo

Waterloo, ON, Canada

Bachelor of Applied Science in Computer Engineering

Expected Apr 2030

- Relevant Coursework: Fundamentals of Programming, Linear Algebra for Engineering, Project Studio

Experience

Autonomous Self Driving Software Engineer

Sept 2025 – Present

WATonomous

Waterloo, ON, Canada

- Coordinated collection and annotation of a comprehensive dataset of 3,448 images (3,087 train, 240 valid, 121 test) for object detection, implementing strategic data augmentation to ensure high-quality training data
- Trained YOLO11s object detection model using Ultralytics framework, achieving a 167% performance boost from 30% to 80% confidence rate through optimized hyperparameters and data augmentation strategies
- Exported trained model to ONNX format with onnxslim optimization for embedded deployment, creating a 36MB artifact ready for real-time inference on rover hardware

Autonomy Developer

Aug 2025 – Present

Waterloo Aerial Robotics Group

Waterloo, ON, Canada

- Developed Hardware-in-the-Loop (HITL) simulation systems in Python to address autonomous flight testing requirements for the 2026 National Annual UAS Student Competition
- Engineered image and GPS synchronization pipeline with panning capabilities to solve real-time sensor data correlation
- Redesigned GPS movement architecture using periodic functions to eliminate unrealistic teleportation behavior, creating smooth trajectory interpolation systems that enhanced simulation realism and accuracy by 75%

Robotics Software Programmer – VEX Robotics Competition

Sept 2024 – May 2025

Checkmate Robotics Club

Markham, Ontario

- Programmed competition software for Team 16868C Rushdown in the VEX Robotics Competition, securing invitation to the 2025 VRC World Championships among top 10% of global teams
- Deployed advanced robotics systems using Okapi PROS Library, Odometry positioning, and PID Controllers, boosting robot performance by 25% through precise motion control and comprehensive error handling
- Designed and refactored autonomous navigation algorithms, creating 15-second match routes and 60-second programming skills routines that achieved 5th place out of 80 teams at Provincial Championships

Projects

Autonomous Vehicle Control System | C++ | Docker | CMake | Foxglove | [Source Code](#) | [Demo Video](#)

- Engineered LiDAR-based perception pipeline processing sensor data to generate real-time costmaps with 0.1m resolution, integrating persistent memory mapping system for 95% accurate dynamic environment representation
- Orchestrated autonomous navigation stack featuring A* path planning algorithm and Pure Pursuit tracking algorithm, achieving sub-0.5m waypoint accuracy and 100% obstacle avoidance success rate across 50+ test scenarios

Fruit Ninja AI | Python | YOLO11 | [Source Code](#)

- Architected and implemented an automated computer vision pipeline using YOLOv11 object detection, achieving 90% accuracy in fruit classification
- Built comprehensive dataset of 816 images (514 training, 302 testing), optimizing model performance through strategic data augmentation and validation techniques

TTC Escape Puzzle Game | C# | Unity | Git | [Source Code](#)

- Designed innovative puzzle game concept for the 2025 GMTK Game Jam, leading a team of 3 developers
- Developed and implemented core gameplay mechanics using Unity and C#, creating a Toronto Transit Commission-inspired puzzle game featuring looping mechanics that enhanced player engagement

Skills

Languages: Java, C, C++, Python, HTML5, CSS, JavaScript, C#, TypeScript

Engineering Tools: Git, Fusion 360, Docker, Foxglove, Unity, Ultralytics

Technologies: ReactJS, OpenCV, Tailwind CSS